EVIDENCE-FIRST PORTFOLIO

AMR / ECSA / FFG Dossier

Predictive AMR architecture and project-management evidence.

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Why ECSA belongs here

ECSA is my future project answer: an AI-for-science direction where uncertainty, governance, validation, and model accountability matter. It should not replace the Secure AI infrastructure proof, but it gives the Fellowship a concrete research direction.

The architecture models antimicrobial resistance as a probabilistic graph/state-space problem with resistance, uncertainty, and stability rather than one static black-box prediction.

- Start with synthetic/public data before clinical data.
- Build Bayesian update behavior and graph learning under explicit uncertainty.
- Add governance, audit logs, model cards, and intended-use boundaries before real-world claims.
- Use lab-in-the-loop validation only after a reproducible proof of concept exists.

POC sequence

Stage	Build	Success criterion
1. Synthetic/public simulator	Small AMR state-space simulator with R, U/sigma, S, transitions, and noisy observations.	Reproducible behavior without private clinical data.
2. Bayesian update layer	Uncertainty-aware updates from synthetic or literature-derived observations.	Calibration and uncertainty shrinkage can be measured.
3. Graph learning layer	Link prediction or graph neural network experiments for missing collateral-sensitivity relationships.	Predictions compared against withheld or simulated links.
4. Governance/audit layer	Model cards, provenance, intended-use boundary, failure modes, and human review gates.	No output can be mistaken for validated clinical advice.
5. External validation plan	Lab/stakeholder protocol before real-data claims.	Independent expert review gates the next step.

Connection to my agent systems

ECSA is not random next to Miguel. The same engineering concerns transfer: model routing, evaluator loops, memory provenance, uncertainty, auditability, and human-in-the-loop control.

A Miguel-like infrastructure could support literature ingestion, hypothesis generation, experiment tracking, model evaluation, and governance documentation, while ECSA gives that infrastructure a high-stakes scientific target.

What it shows

AMR-HORIZONT is project-management and governance evidence: a 12-month plan with work packages, stakeholder strategy, budget logic, outputs, Q&A, red-flag handling, and scenario testing.

It should be presented as planning/submission evidence, not as funded execution unless a separate official confirmation is provided.

Project-management signal

- 12-month project framing with work packages and outputs.
- Stakeholder segmentation and participatory One-Health logic.
- Budget, external expertise, evaluator Q&A, and red-flag handling.
- Scenario-based testing without claiming clinical or patient-data validation.

How I would explain it

I include AMR-HORIZONT because it shows that I can turn a complex health problem into a structured project package. It is not code-first evidence; it is planning, governance, stakeholder and execution evidence.

For Fellowship, I would keep it secondary. For the general portfolio, it matters because it shows I can organize beyond my own machine and think in work packages, risk, outputs and evaluation.